

Model 44

A rack channel strip in the spirit of the early-90s cassette multitrackers: clean preamp, semi-parametric EQ and tape.



Overview

Model 44 is a single channel strip in the spirit of the early-90s rack-mount cassette multitrack gear. The signal goes through a clean, high-headroom preamp, a three-band EQ with a sweepable mid, and then a tape stage: a dense magnetic saturation ceiling with a NORMAL / DOUBLE speed switch and compander noise reduction.

The preamp itself stays clean - the colour and the ceiling come from the tape. Double speed raises that ceiling, extends the top end and tightens the sub-bass, for a punchier, cleaner sound; normal speed is softer, darker and saturates earlier. The noise reduction is tuned to pump less than the earlier 80s machines.

Reach for it for

- Adding cassette-tape density and warmth to a single track
- Drums and bass that need to sit tighter and punchier (DOUBLE speed)
- Quick tone shaping with a sweepable mid
- A softer, darker lo-fi tone (NORMAL speed)
- Parallel tape colour with MIX

Controls

CHANNEL

Signal light	The green light glows when audio is coming out.
GAIN	The preamp gain, -6 to +24 dB (+6 dB by default). The preamp itself stays clean; turning it up drives the tape stage harder.
MID FREQ	The centre of the sweepable mid band, 200 Hz to 5 kHz.
MID GAIN	Cut or boost at MID FREQ, -12 to +12 dB, with a fixed, fairly broad width.
TREBLE	A high shelf at 12 kHz, -10 to +10 dB.
BASS	A low shelf at 80 Hz, -10 to +10 dB.
TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	On by default. GUARD gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light shows when it is on.
MIX	Blends the coloured signal with the dry input, 0 to 100%. 100% is fully through the strip.

Meters

L / R	VU meters showing the left and right output.
--------------	--

TAPE

DRIVE	How hard the tape's saturation ceiling works, 0 to 100% (35% by default). More DRIVE rounds the peaks off sooner and harder, for a denser sound.
--------------	--

NORMAL / DOUBLE	The tape SPEED. NORMAL (1 7/8 IPS): the top end rolls off around 10 kHz and the saturation starts earlier. DOUBLE (3 3/4 IPS): the ceiling sits higher, the top end reaches about 16 kHz, the sub-bass below about 45 Hz is tightened and transients stay punchier.
NR	Companer noise reduction, in by default. It compresses the signal 2:1 with a treble lift on the way onto the tape and expands it 1:2 coming back. Because the expander follows what comes off the tape, it breathes a little, though more gently than the 80s units. Switch it off for the raw tape sound.
OUTPUT	The output level, -24 to +12 dB.

The header

< / >	Step through the presets.
PRESETS	Opens the list of factory presets. Choosing one sets the CHANNEL EQ, GAIN, the TAPE section and MIX; OUTPUT, TRIM and GUARD are left as they are. The last entry resets everything to the defaults.
PATCH	Shows the name of the preset last chosen.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Output stage

These settings have no knobs on the panel; reach them through REAPER's parameter list for the plugin (or automation lanes).

Band sat	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
-----------------	---

Bias	Shapes the saturation: more bias adds more even (2nd) harmonics, for a warmer, rounder colour.
Sat low band / Sat high band	Where the low and high saturation bands start (40 - 400 Hz and 2 - 16 kHz).
Ceiling / limit threshold	With Clip: the level the peaks are softly clipped at. With Limit: the threshold of a brickwall limiter that holds the peaks at -0.3 dBFS - pulling it down limits harder and makes the output louder.
Output stage	Off, Clip or Limit: nothing, a soft clipper (no latency, the default), or a lookahead limiter (3 ms latency, compensated by REAPER).

Presets

Clean Double Speed	Double speed with light drive, a small treble lift and NR in.
Normal Speed Warm	Normal speed, more gain, a little low-mid and bass lift, NR in.
Rack Punch	Double speed, 14 dB of gain and 55% drive, with a 2.5 kHz and treble lift.
Scooped Mids	Double speed with a 4 dB cut at 600 Hz and a touch of treble and bass.
No NR	Normal speed with the noise reduction out: the raw tape.
Reset all to defaults	Every control back to its starting value, including OUTPUT, TRIM, GUARD and the output stage.

Tips

- Set GAIN for the amount of tape colour you want, then bring OUTPUT down to match the level before you judge it.
- Switch between NORMAL and DOUBLE on drums: double speed keeps the snap, normal speed softens and thickens it.

- With NR in and plenty of DRIVE you can hear the compander breathe; switch NR out if you want the saturation without the movement.
- Use MIX at 30-60% to add tape density under a track without losing its transients.

File: gfx-model-44.jsfx